AfCEN BUILD 2026 · NIGERIA

Nobody in the room speaks both languages.

A two-way sign-language interpreter that fits in a nurse's pocket.

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THE WARD, TODAY

The interpreter is not in the room.

1. An NSL interpreter has to be booked. In practice that means days — and often an interpreter in another state.

2. So the encounter is carried by a relative, or by a pen and a scrap of paper.

3. The three things that must never be guessed — where it hurts, what the patient is allergic to, and whether they consent — are exactly the three that get guessed.

4. The Discrimination Against Persons with Disabilities (Prohibition) Act 2018 already requires effective communication in public services.

THE MEASUREMENT

Nigeria publishes no ratio of Deaf patients to qualified interpreters. The gap is not measured, which is part of why it does not

THE PRODUCT

Two directions. One screen.

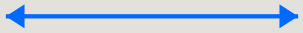
PATIENT → CLINICIAN

The patient signs into the camera.

Hand shapes are read on the device.

English appears on the clinician's screen, and is spoken aloud.

BOTH WAYS



CLINICIAN → PATIENT

The clinician speaks.

Speech becomes text in the browser.

The reply is played back to the patient in sign.

Runs in the browser of the phone the nurse already carries. Nothing to install. No account. Survives ward wifi that drops after first load.

QUALITY OF AI · 25%

Three systems, one rule: never guess into a decision.

SPEECH → TEXT

The clinician speaks. Browser speech recognition turns it into text. No cloud round trip.

HAND SHAPE → MEANING

The patient signs into the camera. A geometric classifier reads a closed set — yes, no, not sure, numerals 1 to 5.

TEXT → SIGN

The reply is returned to the patient in Nigerian Sign Language, built on licensed engines we pay for.

We do not claim to train a sign-language model. The AI we own is the layer that decides what is understood, what is refused, and what the patient has to confirm before it counts.

Speech recognition and hand tracking run in the browser. Nothing a patient signs leaves the device in this build.

STATUS · TRACTION 20%

What works today. What does not.

EXISTS

— Live camera capture and hand tracking

— Reads yes, no, not sure, and the numerals 1 to 5

— Clinician speech to text; the patient's answer spoken aloud

— Refuses out loud when a phrase falls outside its set

— Runs in a browser, on ward wifi

DOES NOT EXIST

— A trained NSL model

— A published accuracy number

— A hospital pilot

— A Deaf patient who has used it

— Revenue

The refusal is the feature. A tool that invents a dosage is worse than one that admits it did not understand.

IMPACT AND REACH · 30%

A law already says yes. Nothing delivers it.

0

paid staff — the first hires are Deaf advisers and NSL interpreters

200

clinical terms is the whole scope. Same structure in every market

0

patient videos stored. Nothing a patient signs leaves the device in this build

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REACH

Deaf Nigerians, in public hospitals. Access is required by law and carried, in practice, by relatives and paper. No account, and no smartphone of the patient's own — it runs on a shared device a clinic already owns.

ACROSS BORDERS

The clinical vocabulary is the product; the sign language is a parameter. Ghana, Kenya and South Africa share the interpreter shortage. Expansion is vocabulary and validation work, not a rebuild.

JOBS

Every new market adds a paid vocabulary team — Deaf advisers and qualified interpreters. That is skilled work that barely exists as a paid role outside a few urban centres, and it stays in the region.

Healthcare is a service that public and private buyers already pay for. We are not asking them to buy something new.

Built so a clinician can see when it is unsure.

DATA

- Conversations are not stored by default
- No patient data trains or fine-tunes a model
- A record, where a hospital asks for one, holds what was understood, the confidence on it, and what the patient confirmed — never raw video
- What leaves the device will be stated plainly, not implied

FAIRNESS AND INCLUSION

- Nigerian Sign Language, built with paid Deaf advisers — not consulted once, employed
- American or British models do not transfer, and we say so
- Runs after first load on ward wifi that drops
- No account, no personal handset, no literacy requirement

Confidence is displayed, never hidden. Outside the vocabulary the app refuses and asks for a human interpreter.

POLICY ALIGNMENT · 10%

The obligation exists. The delivery does not.

01 **The 2018 Disability Act**

Requires effective communication in public services, health included. Unmute is the mechanism, not a new obligation.

02 **NITDA's digital economy agenda**

Local AI capability, applied to a public service, built by a Nigerian founder.

03 **Digital services that buyers rely on**

Hospitals already pay for interpreters and for paper processes. This is a service line, not a subsidy.

Aligned with the AfCEN Build theme: AI in the services that public and private buyers depend on, creating skilled work in the region.

TEAM

One founder. The gap is the team, and I know it.

Prince Nzubechukwu Michael — founder, sole team member.

Product, the clinical vocabulary scope, and the build. Built the prototype from a phone, without a laptop, using AI-assisted development. A year inside this problem: mapping the vendor landscape in both directions, and defining the clinical vocabulary that carries the risk in a routine encounter.

No engineering team yet. The first hires are Deaf advisers and qualified NSL interpreters, not engineers — the model is bought, the vocabulary cannot be.

FIRST HIRES

Deaf advisers

NSL interpreters

THE ASK

Three things, none of them a cheque.

01 **Two pilot departments**

One general hospital and one community clinic in the FCT. Free for six months, in exchange for ward access, consented sessions, and the vocabulary that actually gets used.

02 **NSL advisers, paid**

Deaf advisers and qualified interpreters to build the two hundred terms. Not consulted once — employed.

03 **The network**

NITDA and Federal Ministry of Health digital health programmes. State hospitals already carry the accessibility obligation.

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